



HABIB UNIVERSITY

Data Structures & Algorithms

CS/CE 102/171 Spring 2025

Instructor: Maria Samad

Date: April 7th, 2025

Quiz # 3

Time Allowed: 60 minutes

Max Points: 40

Name: _____

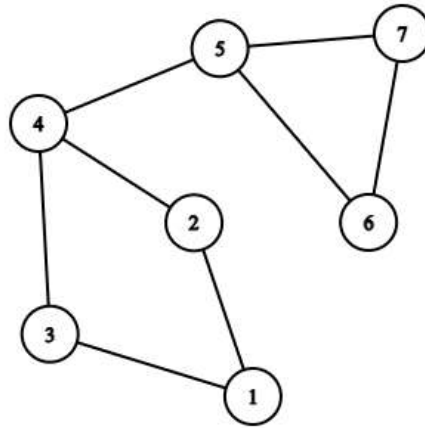
QUESTION 1: HASHING (15 marks)

- A hash table contains student IDs (integers) as keys, and their names as values. Write an algorithm using Hashing that gets the value, i.e., name of the student from the Hash table when the key, i.e., student ID is provided. The key will be taken as input from the user. Assume N to be the size of existing hash table. Hash function to be used is Divide Modulo, and collisions must be handled using Key Offset Rehashing with the Secondary Hash function defined as: $(id * 2) + 2 \text{ mod size}$. If the key is not found, return a message stating “**Not Found**”, otherwise return the value. You may use two separate lists to represent keys and values just like you did in the labs. **YOU ARE NOT ALLOWED TO USE ANY ADDITIONAL DATA STRUCTURES, APART FROM THE ONES SPECIFIED IN THE QUESTION.**
- For the given hash table, dry run the algorithm from part (a), to get the values for the student ID = 85. Don't just write down the answer by looking at the hash table. Show each step to get the value. Assume size of hash table is 9.
Keys = [99, None, 24, None, 49, 40, 15, 33, 85]
Values = [“Faiz”, None, “Bilal”, None, “Dania”, “Zeeana”, “Ariana”, “Emaad”, “Abdul Ahad”]

QUESTION 2: GRAPH TRAVERSAL (10 marks)

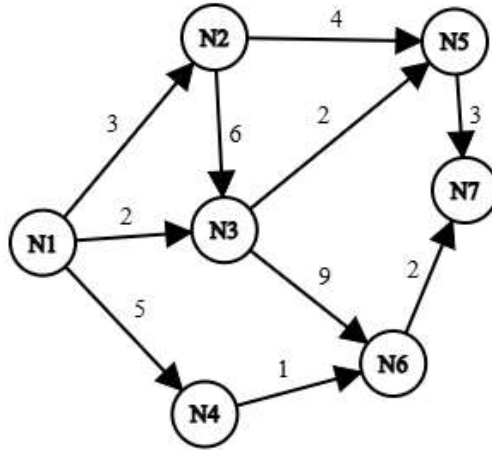
For the given graph, apply the Depth-First Search Traversal Algorithm with the starting vertex = 6. Show all the working for each step. Construct the corresponding DFS graph, by showing all the tree edges and non-tree edges. Use different notation to represent each edge, and specify a legend for it. Follow ascending order in choosing vertices. Calculate the total cost of traversal in the end.

IF YOU ARE MISSING ANY INTERIM CALCULATIONS, YOU WILL NOT GET ANY MARKS



QUESTION 3: SHORTEST PATH (15 marks)

- a. Run the Dijkstra's Algorithm on the following directed, weighted graph, to get the shortest path of every vertex from the starting vertex, N1. Specify shortest path of every vertex from the starting vertex, and calculate the cost of this shortest path. Draw the reduced graph after Dijkstra's algorithm has successfully been implemented, showing the shortest paths between all the vertices. Show your complete working. **IF YOU DO NOT SHOW ANY WORKING, YOU WILL NOT GET MARKS FOR IT**



- b. For the same graph, as given in part (a), define the Graph using Edge List Representation

